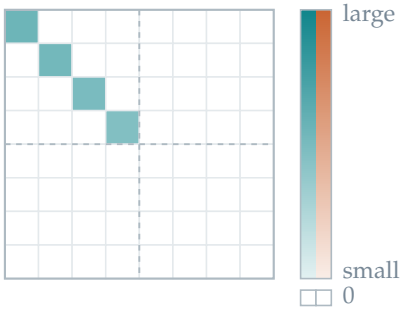


a Shot noise lifts the rank

Singular values (schematic)

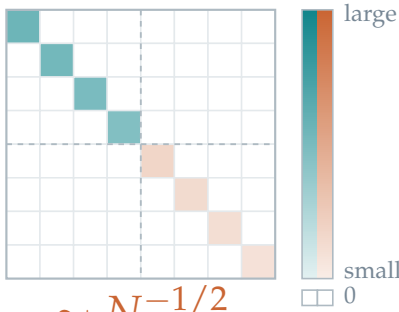
Ideal probabilities



$\text{rank}(P) \leq d^2$

shots

Finite-shot features

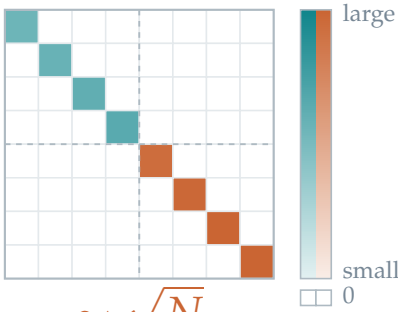


$\sim N^{-1/2}$

New noise-only directions

invert

Pseudoinverse



$\sim \sqrt{N}$

Small modes, large inverse

Exact targets + exact test
Divergence cancels

Noisy targets + finite test
Quantum overfitting

b Origins of the MSE

Training-shot effects



Training bias

Training shot noise

N^{-2}



Training fluctuations

Training shot noise

$1 / (N n_{\text{tr}})$

Target noise
Classical



Training target noise

Noisy training labels

γ^2 / n_{tr}



Test target noise

Noisy test labels

γ^2

Test-shot effect



Test sampling

Finite test shots

$1 / M$

Quantum overfitting



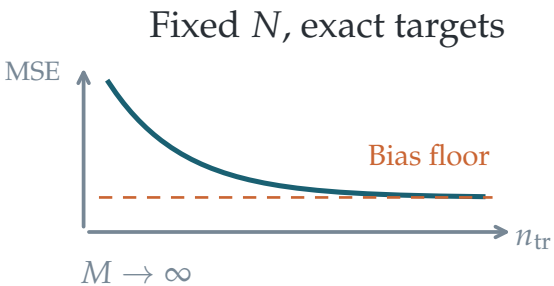
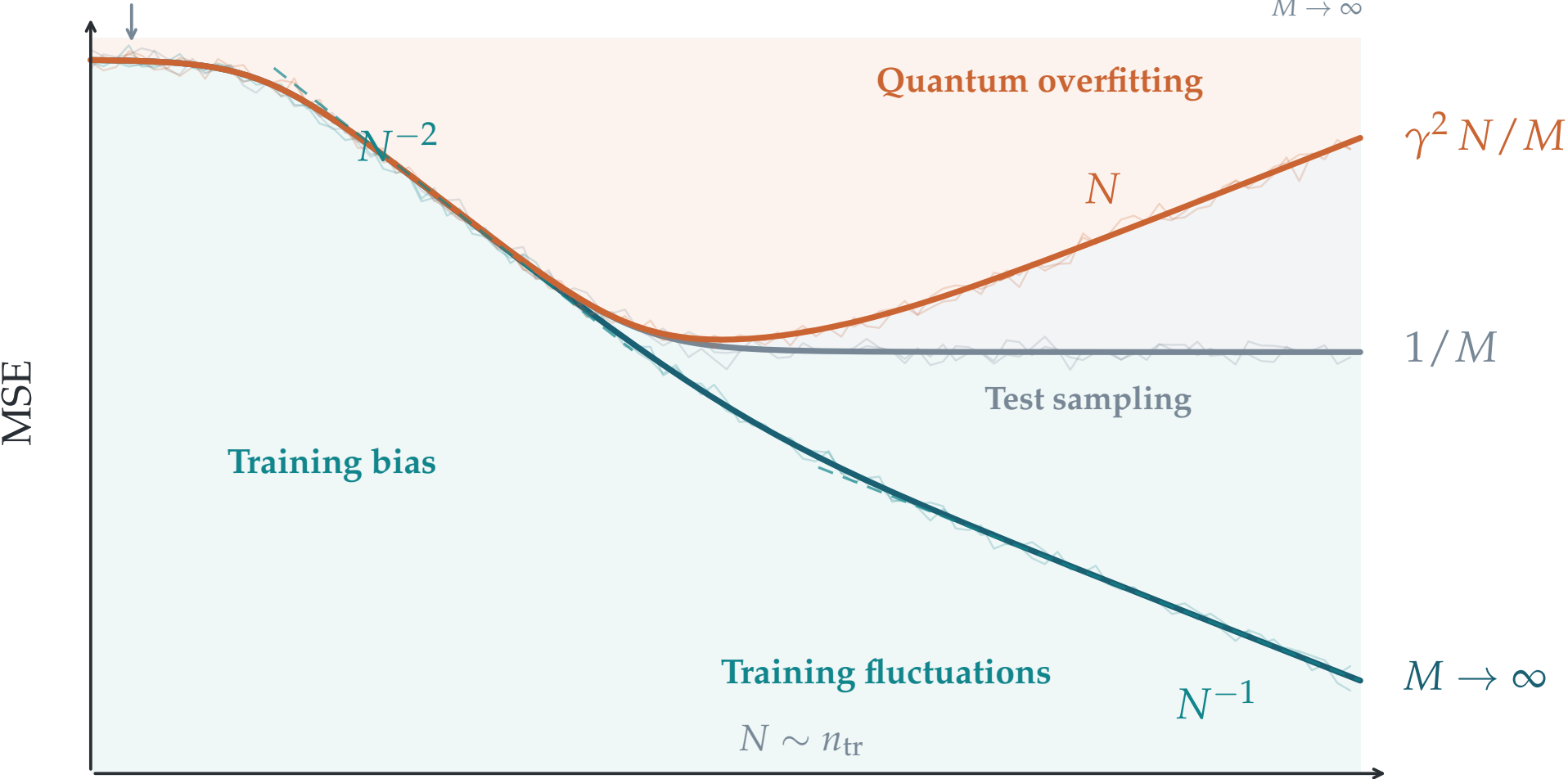
Quantum overfitting

Training-shot noise $\times$ target noise
 $\times$ test-shot noise

$\gamma^2 N / M$

c Resources shape the MSE

Undersampled



Training shots per state, N